

Alfido Tech Internship

Task 3: Data Logger – Sensor to SD Card

Name:

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Domain:

Embedded Systems

Tools Used:

- Wokwi Simulator
 - Arduino IDE
-

1. Objective

The objective of this project is to collect temperature and light sensor data and log the readings for monitoring and analysis. The system demonstrates sensor interfacing, data acquisition, and data logging using Arduino.

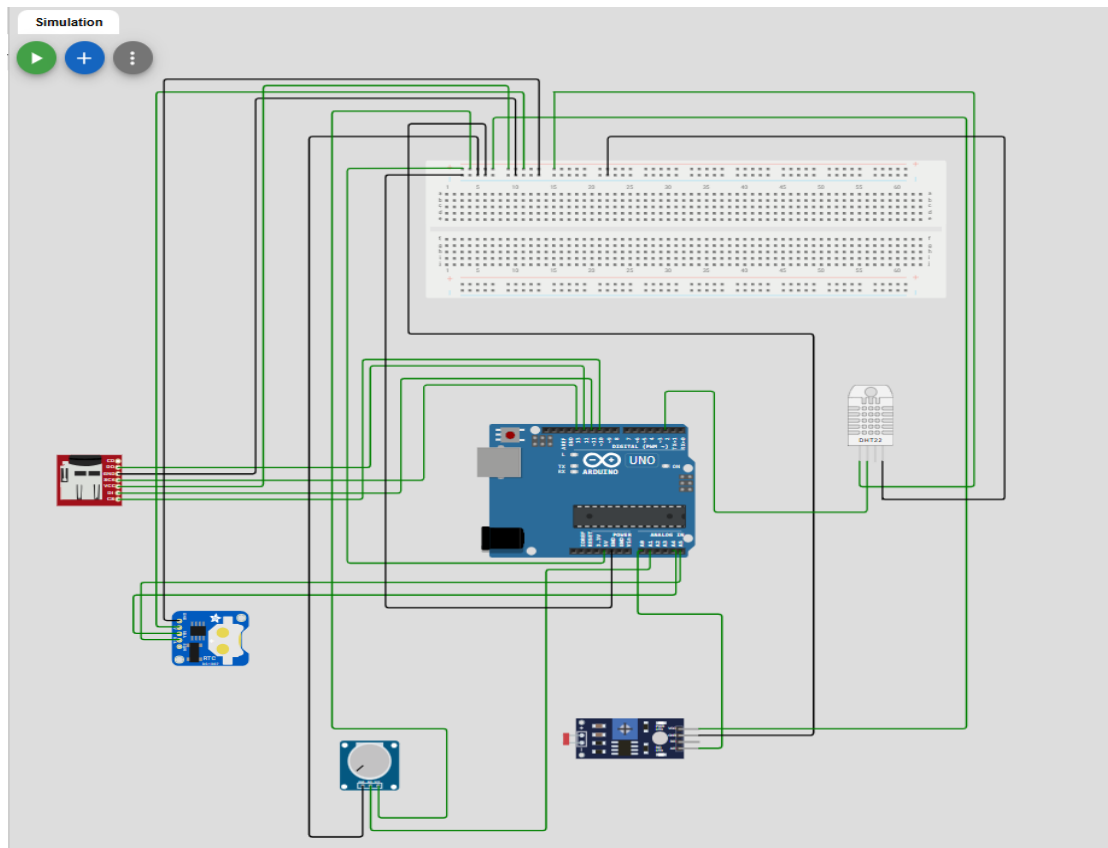
2. Components Used

- Arduino UNO
 - DHT22 Temperature Sensor
 - LDR Sensor Module
 - SD Card Module
 - RTC Module (DS3231)
 - Potentiometer
 - Breadboard
 - Jumper Wires
-

3. Software Used

- Wokwi Simulator
 - Arduino IDE
-

4. Circuit Diagram



5. Working Principle

The DHT22 sensor measures temperature and the LDR sensor module measures light intensity. Arduino continuously reads the sensor values and generates data samples in CSV format. The RTC module can be used to provide timestamps for recorded data, while the SD card module can be used to store the collected information. In the simulation environment, the collected data is displayed through the Serial Monitor in CSV format.

6. Arduino Code

```
#include <DHT.h>
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT22
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
const int lightPin = A0;
```

```
int sampleNumber = 0;

void setup() {
  Serial.begin(9600);
  dht.begin();

  // CSV Header
  Serial.println("Sample,Temperature_C,Light_Value");
}

void loop() {

  float temperature = dht.readTemperature();

  int lightValue = analogRead(lightPin);

  sampleNumber++;

  // CSV Format Output
  Serial.print(sampleNumber);
  Serial.print(",");

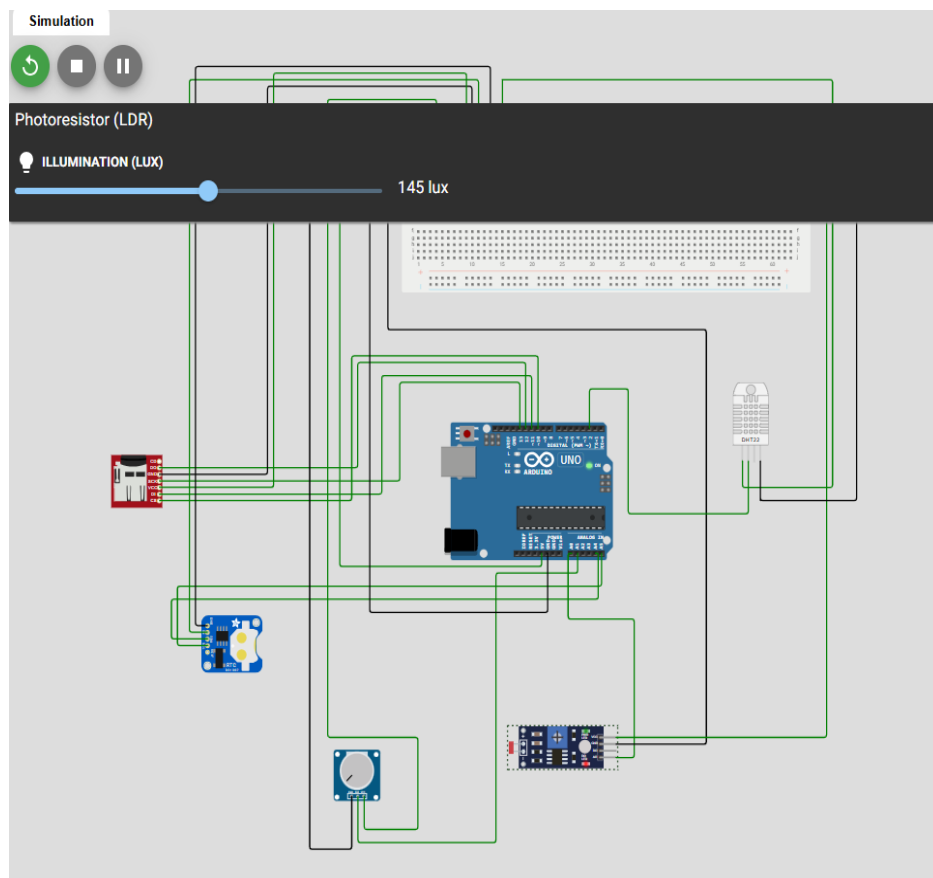
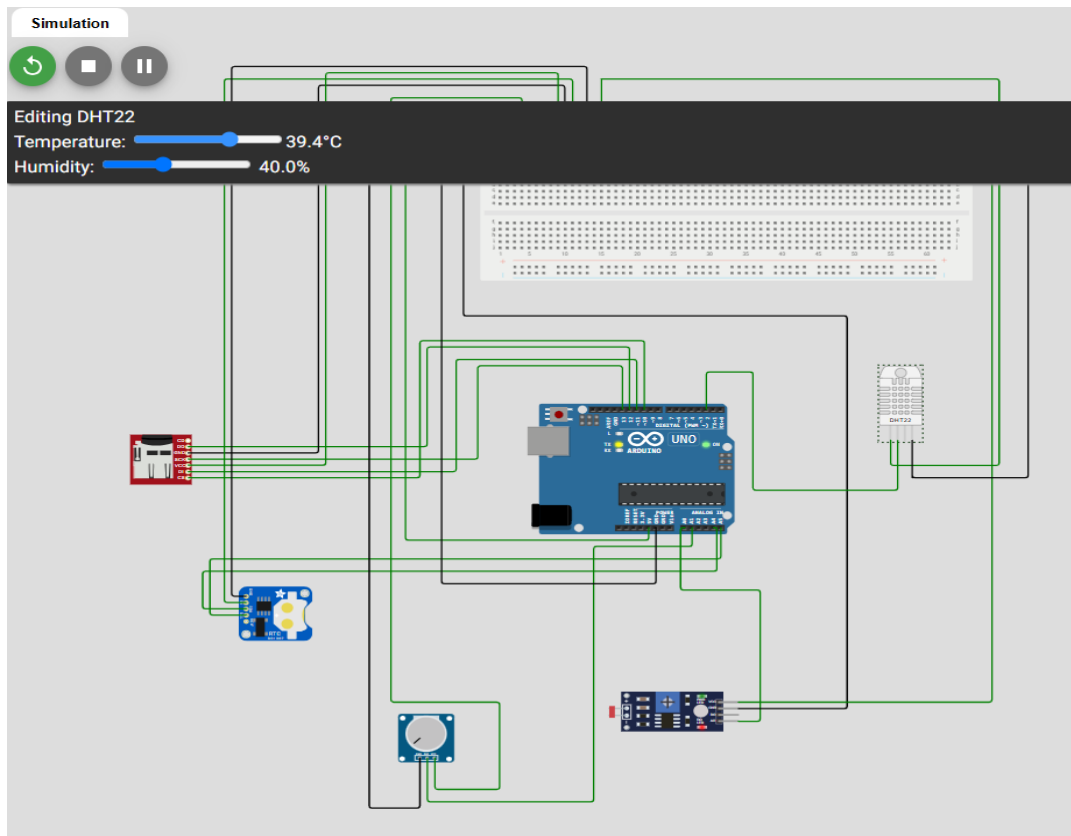
  Serial.print(temperature);
  Serial.print(",");

  Serial.println(lightValue);

  delay(1000); // 1 sample per second
}
```

7. Output Screenshots

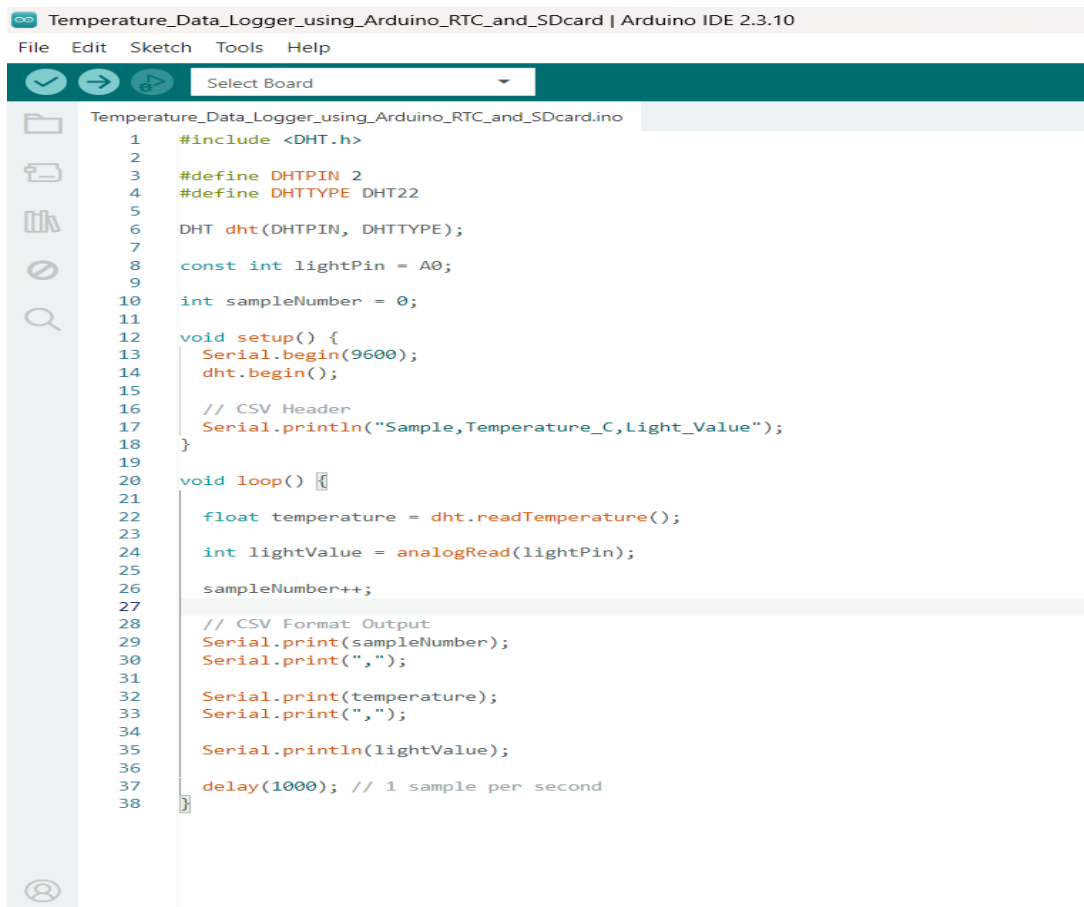
Circuit Screenshot



Serial Monitor Output and CSV data output

```
Sample, Temperature_C, Light_Value
1, 22.50, 294
2, 22.50, 294
3, 22.50, 294
4, 22.50, 294
5, 22.50, 294
6, 22.50, 294
7, 33.10, 294
8, 33.10, 294
9, 33.10, 294
10, 33.10, 165
11, 33.10, 165
12, 33.10, 165
13, 33.10, 165
14, 33.10, 165
15, 33.10, 165
16, 33.10, 165
17, 33.10, 165
18, 33.10, 165
19, 21.40, 165
20, 21.40, 165
```

Code Screenshot



```
Temperature_Data_Logger_using_Arduino_RTC_and_SDcard | Arduino IDE 2.3.10
File Edit Sketch Tools Help

Select Board

Temperature_Data_Logger_using_Arduino_RTC_and_SDcard.ino
1  #include <DHT.h>
2
3  #define DHTPIN 2
4  #define DHTTYPE DHT22
5
6  DHT dht(DHTPIN, DHTTYPE);
7
8  const int lightPin = A0;
9
10 int sampleNumber = 0;
11
12 void setup() {
13   Serial.begin(9600);
14   dht.begin();
15
16   // CSV Header
17   Serial.println("Sample, Temperature_C, Light_Value");
18 }
19
20 void loop() {
21
22   float temperature = dht.readTemperature();
23
24   int lightValue = analogRead(lightPin);
25
26   sampleNumber++;
27
28   // CSV Format Output
29   Serial.print(sampleNumber);
30   Serial.print(",");
31
32   Serial.print(temperature);
33   Serial.print(",");
34
35   Serial.println(lightValue);
36
37   delay(1000); // 1 sample per second
38 }
```

8. Results

The system successfully collected temperature and light sensor readings and displayed the data in CSV format through the Serial Monitor. The generated data samples demonstrated the functionality of a basic data logging system and can be extended for SD card storage in real hardware implementations.

9. Conclusion

This project demonstrates the fundamental concepts of embedded systems including sensor interfacing, data acquisition, serial communication, and data logging. The system successfully monitored environmental parameters and generated structured sensor data suitable for storage and analysis.

10. GitHub Repository Link

[redypraveena239-prog/Task-3-Data-Logger: Arduino Data Logger using DHT22 Temperature Sensor and LDR Sensor](https://github.com/reddypraveena239-prog/Task-3-Data-Logger: Arduino Data Logger using DHT22 Temperature Sensor and LDR Sensor)

11. Demo Video Link

https://drive.google.com/file/d/1wElw7HRz64R_U_cHYo60h7t4nUwPX1JA/view?usp=sharing